

Public Economics (ECON 131)
Section #3: Income Inequality and Tax Incidence

GSI: Steve Ramos

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1 Difference-in-Differences

Difference-in-Differences (DiD or diff-in-diff) is an important tool economists use to conduct credible quasi-experiments. We will see many examples of papers using the DiD design. It is important to know how to answer questions on empirical effects quickly, concisely, and (most importantly) correctly. If you are asked about a *causal* effect, and you see a graph with two time series and policy break, you should already be thinking about DiD!

1.1 Checks and Implementation

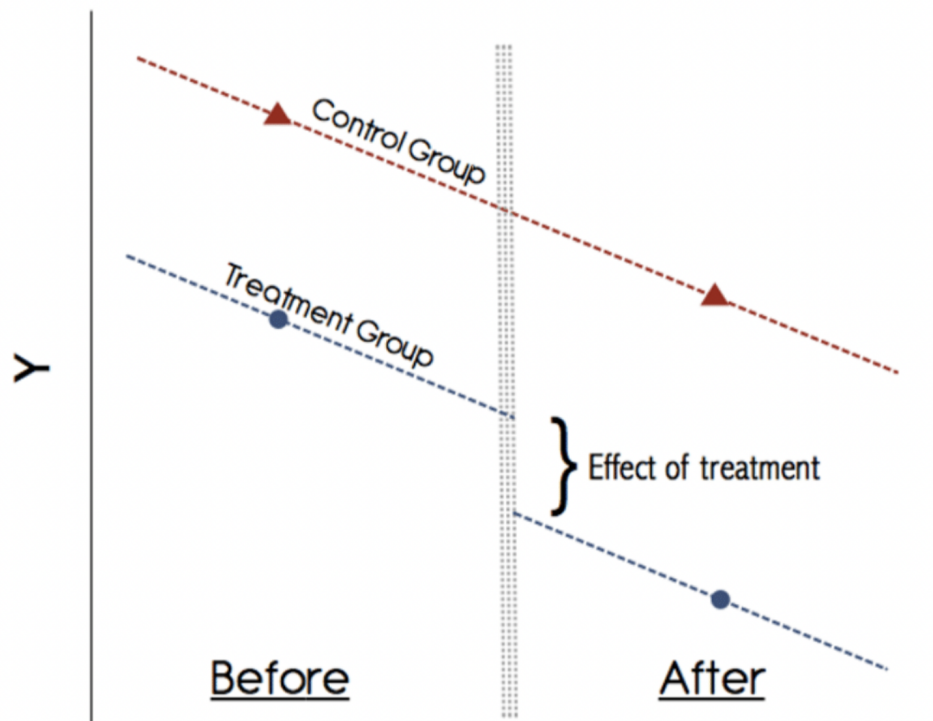
For concreteness, let us consider the following (fictional) setting. In June 2020, Berkeley decided to raise gas taxes on cars with license plates ending in odd numbers, with no change to cars with even-numbered license plates. We will call cars with odd-numbered plates the "treatment" group, and those with even numbered plates the "control" group (since only the former is affected by the policy). Armed with weekly driving data from the city of Berkeley, you have decided to evaluate the policy: do these taxes lead to less driving? We will analyze potential graphs below.

To analyze a policy using DiD, you need to follow these steps:

1. Do the treatment and control time series follow a parallel trend before the tax is enacted? That is, if you take the difference between the treatment and control time series, is it the same over time? If yes, calculate this difference.
 - (a) If yes, then you can apply the DiD design to say something about the **causal effect** of the tax.
 - (b) *Note 1:* When using DiD, we are assuming that in the absence of the treatment (in this case, the tax) that the treatment and control groups would have evolved similarly before and after the policy. This assumption is referred to as **parallel trends**. We typically test this assumption graphically. If the trends are not similar in the pre-policy period, then it is unlikely the control group is a good counterfactual for the treatment group. In other words, the parallel trends assumption would be violated and we can not apply DiD.
 - (c) *Note 2:* The important part here is to analyze the trends **before** the policy. The trend afterwards is affected by the policy!
 - (d) *Note 3:* There are other assumptions used for DiD, but for the purposes of this class you can take them as given. If you are interested in knowing more about these, come see me in office hours.
 - (e) *Note 4:* If you are interested in how DiD lines up with regression, see the appendix.
2. Does the treatment group time series have a trend break **after** the policy is implemented? If yes, calculate the difference between the treatment and control time series after the policy is implemented.
 - (a) *Note (very important) :* by itself, this difference does **NOT** give us the causal effect of the policy! This is because there may have been a pre-period difference between the two groups, which we will need to deal with as well.

3. Take the difference we calculated in (2), and subtract it from the difference calculated in (1): this gives you the **causal effect**.
 - (a) That is why we call it difference-in-differences!
 - (b) Note 1: This is a causal difference!
 - (c) Note 2: If (1) was satisfied, but the difference-in-differences estimate is extremely small and/or approximately zero, this still means you have a causal effect: it is just that the causal effect is zero.

Figure 1: A valid DiD example.



Things to note on the figure above:

- Both lines are trending down over time. This is not a problem for DiD!
- The treatment group is always lower than the control group. This is also not a problem for DiD!
- The most intuitive way for me to think about the DiD estimated effect is to visually do the following:
 - Take the difference in the pre and post periods separately.
 - Take the difference between these differences.

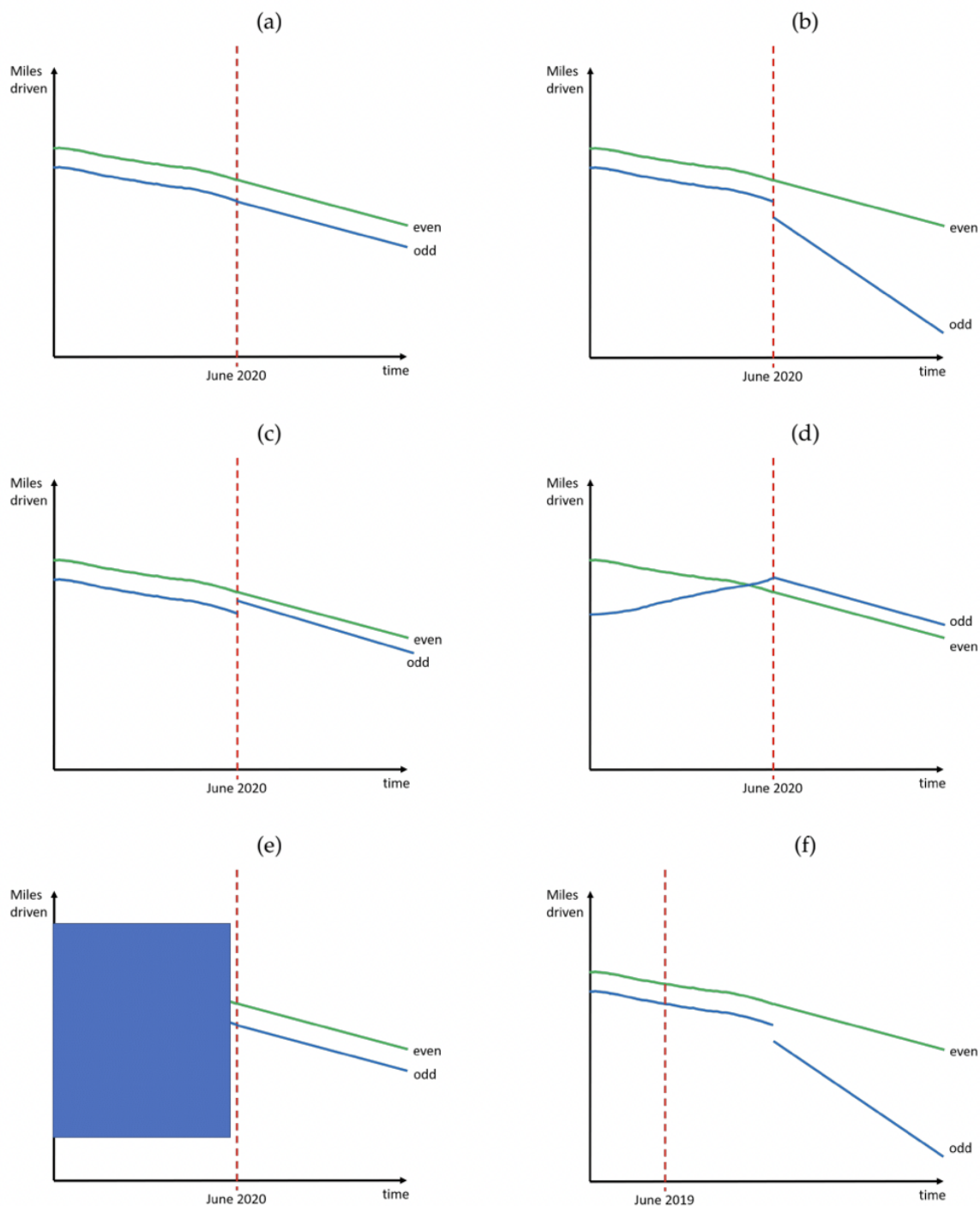
We can also do this in "table" form, which requires the means for each group and period. The table below (reproducing a famous 1994 paper) reports the average number of hours worked at restaurants on the border of New Jersey (which had a minimum wage change) and Pennsylvania (which did not), reported in February and November:

	New Jersey	Pennsylvania	Difference
February	20.44	23.33	-2.89
November	21.03	21.17	-0.14
Difference	0.59	-2.16	2.75

The important thing to remember is that you need **both differences** to get the causal estimate.

1.2 Example: Gas Taxes and Driving

Figure 2: Different potential DiD plots.



Can you identify the causal effect from each graph (positive, negative, no effect, unable to estimate using DiD)? [Answers, in order: no, negative, positive, unable, unable, no]

2 Tax Incidence

2.1 Definitions

- **Statutory incidence** has to do with who the tax is levied on
- **Economic incidence** has to do with how the burden of the tax is shared
- **Elasticity**: Percentage change in quantity when price changes by one percent. We often use elasticities in economic analysis because they are unit-free.
- **Price elasticity of demand**: $\varepsilon_D = \frac{p}{D} \frac{dD}{dp} = \frac{pD'(p)}{D(p)} < 0$
- **Price elasticity of supply**: $\varepsilon_S = \frac{p}{S} \frac{dS}{dp} = \frac{pS'(p)}{S(p)} > 0$

2.2 Calculating Tax Incidence

- Consumers face price z for good x . Their demand is $D(z)$ which is decreasing in z .
- Producer receive price p for good x . Their supply is $S(p)$, which is increasing in p .
- In equilibrium, prices have to be such that producers want to sell exactly the same amount that consumers want to buy. This is the equilibrium condition: $Q = S(p) = D(z)$
 - When there is no tax, $p = z$
 - If a tax t is leveraged on the consumer, the relationship between consumer and producer prices is $z = p + t$

Draw a graph to visually compare the equilibrium with no tax to that with some tax t levied on the consumer. How would this equilibrium change if demand were perfectly inelastic?

Now for some math:

- Starting from when tax $t = 0$ (so note $z = p$) and $S(p) = D(p)$, we want to characterize dp/dt : the effect of a tax increase on the equilibrium price. The change in the equilibrium price determines who bears the effective burden of a tax.
- A small change in the tax, dt , generates a small price change, dp , so that equilibrium holds:

$$\begin{aligned}S(p + dp) &= D(p + dp + dt) \\S(p) + S'(p)dp &= D(p) + D'(p)(dp + dt) \\S'(p)dp &= D'(p)(dp + dt) \\\frac{dp}{dt} &= \frac{D'(p)}{S'(p) - D'(p)}\end{aligned}$$

- Then, we have the following formulas summarizing responses to taxes:

$$\begin{aligned}\frac{dp}{dt} &= \frac{D'(p)}{S'(p) - D'(p)} = \frac{\varepsilon_D}{\varepsilon_S - \varepsilon_D} \\\frac{dz}{dt} &= \frac{\varepsilon_S}{\varepsilon_S - \varepsilon_D} \\\frac{dz}{dt} &= 1 + \frac{dp}{dt}\end{aligned}$$

- Also, $-1 \leq \frac{dp}{dt} \leq 0$ and $0 \leq \frac{dz}{dt} \leq 1$.

When do consumers bear the entire burden of the tax? ($dp/dt = 0$ and $dz/dt = 1$)

1. $\varepsilon_D = 0$ [inelastic demand]

- Example: short-run demand for gas inelastic (need to drive to work)

2. $\varepsilon_S = \infty$ [perfectly elastic supply]

- Example: perfectly competitive industry

When do producers bear the entire burden of the tax? ($dp/dt = -1$ and $dz/dt = 0$)

1. $\varepsilon_S = 0$ [inelastic supply]

- Example: fixed quantity supplied (land)

2. $\varepsilon_D = -\infty$ [perfectly elastic demand]

- Example: there is a close substitute, and demand shifts to this substitute if price changes.

2.3 Key results to keep in mind

1. The *statutory incidence* of a tax is **not generally equal** to its *economic incidence*.
2. The equilibrium outcome is **independent** of who nominally pays the tax; it does not matter if the statutory incidence is on the buyer or the seller.
3. The **more inelastic** side bears more of the tax burden.

2.4 Practice problems (do at home!)

2.4.1 Gruber, Ch.19, Q.2 (modified)

The demand for rutabagas is $Q = 2,000 - 100P$ and the supply of rutabagas is $Q = -100 + 200P$.

1. Who bears the statutory incidence of a \$2 per unit sales tax on rutabagas, paid by the producer and reflected in the sticker price?
2. Who bears the economic incidence of this tax? (Discuss)
3. What is the equilibrium price without the tax?
4. What is the equilibrium price with the tax?
5. What are the tax burdens on the consumer and producer?
Hint: The consumers' tax burden = (posttax price - pretax price) + tax payments by consumers
The producers' tax burden = (pretax price - posttax price) + tax payments by producers

Solution:

1. Producers bear the statutory incidence, because producers are the ones who have to pay the tax according to the question.
2. Economic incidence is determined by relative elasticities. As we will see, the quantity supplied is more responsive to a change in price, so the less elastic consumers will bear most of the economic incidence. To calculate the relative burdens, solve the equilibrium condition with and without the tax.
3. Without the tax: $2,000 - 100P = -100 + 200P$. Price = \$7.00.
4. With the tax, the price the supplier receives is reduced by \$2.00. The equilibrium condition:

$$2,000 - 100P = 200(P - 2) - 100$$

$$2,000 - 100P = 200P - 500$$

$$2,500 = 300P$$

$$P = 8.33$$

Thus price = \$8.33.

5. The consumers' tax burden = (posttax price - pretax price) + tax payments by consumers, here $\$8.33 - \$7.00 + 0 \approx \$1.33$.
The producers' tax burden = (pretax price - posttax price) + tax payments by producers, here $\$7.00 - \$8.33 + \$2.00 \approx \0.67 . In this case the consumer bears a larger share of the tax burden than the producer.

2.4.2 Gruber, Ch.19, Q.3 (modified)

The demand for rutabagas is still $Q = 2,000 - 100P$ and the supply is still $Q = -100 + 200P$, as in Question 2. Governor Sloop decides that instead of imposing the \$2 sales tax described in Question 2, the government will instead force consumers to pay the tax, such that the tax is no longer reflected in the sticker price. What will happen to the “sticker price” on rutabagas? How will the size of the consumer tax burden change?

Solution:

- The sticker price for consumers when they bear the statutory burden of the \$2 tax is $P \approx \$6.33$. Similarly to question 2, this is the solution to $2,000 - 100(P + 2) = -100 + 200P$.
- Before, the sticker price for consumers when firms pay the tax is the solution to $2,000 - 100P' = -100 + 200(P' - 2)$, so the sticker price in that case is $P' \approx \$8.33$, or \$2.00 more than the sticker price when consumers pay the tax.
- Consumers pay exactly the same net amount as before: now, they pay the \$6.33 sticker price plus a \$2 tax, and before they paid \$8.33 directly. The economic incidence of the tax is unchanged.

2.4.3 EXTRA PRACTICE: Gruber, Ch.19, Q.4

The demand for football tickets is $Q = 360 - 10P$ and the supply of football tickets is $Q = 20P$. Calculate the gross price paid by consumers after a per-ticket tax of \$4. Calculate the after-tax price received by ticket sellers.

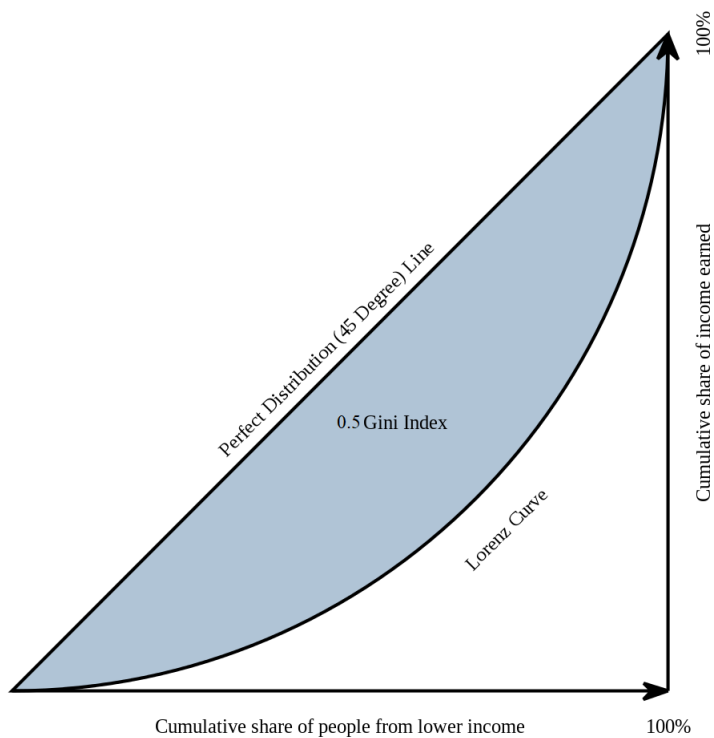
Solution:

- For this answer, it does not matter whether the tax is added to the price paid by the consumers or subtracted from the price the sellers keep.
- Adding it to the consumers' price yields demand of $360 - 10(P + 4)$, which is set equal to supply to yield the equilibrium after-tax price: $360 - 10P - 40 = 20P$.
- Simplified, $320 = 30P$, so base price = \$10.67 and price + tax = \$14.67 paid by consumers.
- Producers keep only $\$14.67 - \$4 = \$10.67$.
- Without the tax, $360 - 10P = 20P \Rightarrow 360 = 30P$; the price paid by consumers and kept by sellers is \$12.

3 Income Distribution

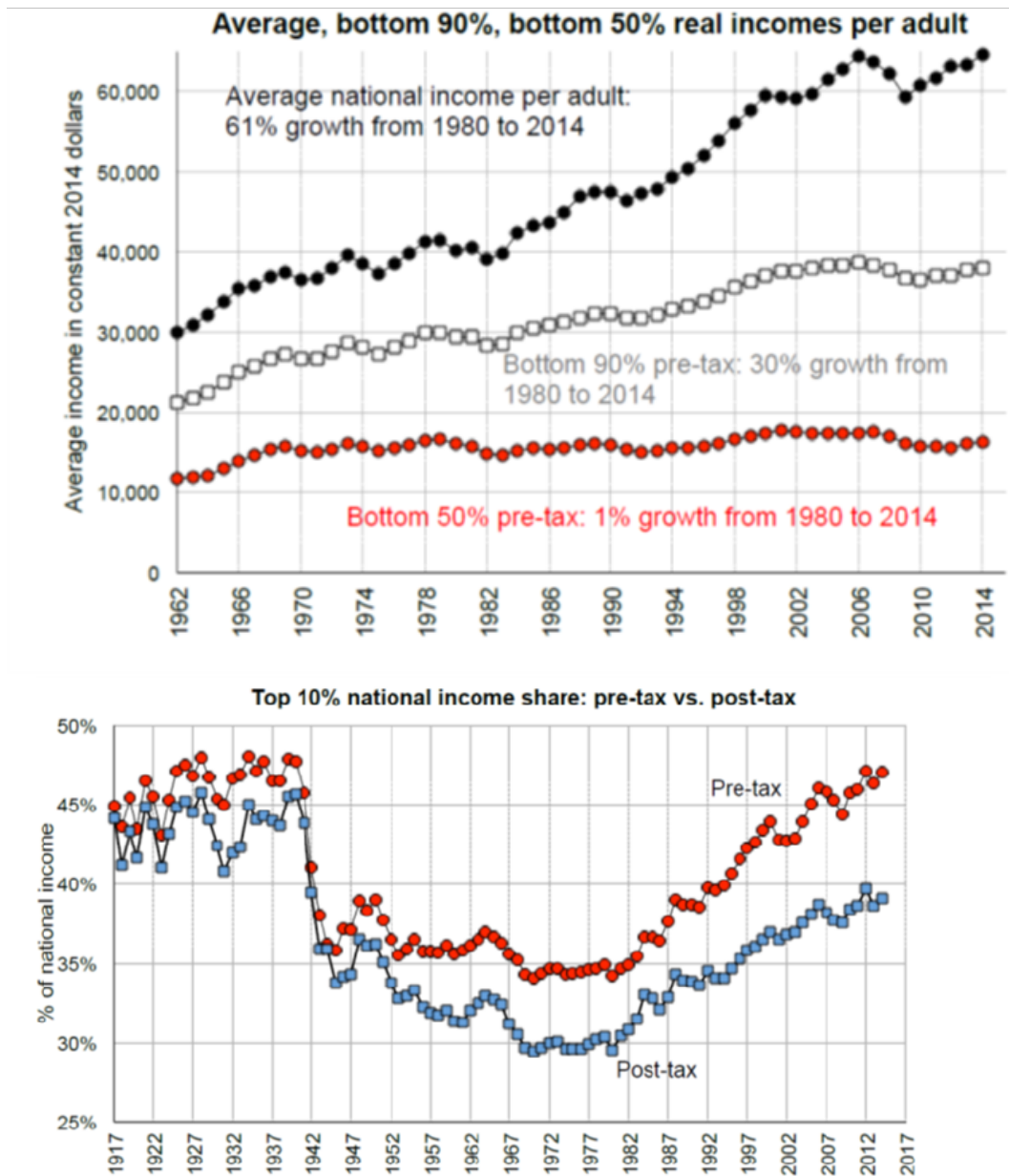
3.1 Measuring Income Inequality

- The **Lorenz curve**, $L(p)$, is a cumulative distribution of income, ie the fraction of total income owned by individuals below the p th percentile of the income distribution.
- The **Gini coefficient** is 2x the area between the 45 degree line and the Lorenz curve.
- The Gini coefficient can be difficult to interpret so, in practice, income inequality is often talked about in terms of shares (eg. the Top 10% of income earners control 40 % of national income)



3.2 Trends in Income Inequality over Time

- Labor and capital income inequality have increased dramatically since the 1980s
- Most of growth in income has come from the top of the income distribution



As you can see, taxes and transfers can be a useful tool for reducing income inequality. To understand the impact of a tax, we have to understand the meaning of elasticities and tax incidence.

4 Appendix: DiD correspondence with regression

Suppose you wanted to run a DiD-style analysis, such as on minimum wage laws in New Jersey and Pennsylvania. You could run the following regression, where *Treatment* is a dummy variable indicating whether someone is in the treatment group, *Post* is an indicator variable for whether the observation comes from after the change. The coefficient that we interpret causally (assuming all our assumptions are correct) is β_3 .

Suppose you have this regression: $Y_i = \beta_0 + \beta_1 \text{Treatment}_i + \beta_2 \text{Post}_i + \beta_3 \text{Treatment}_i * \text{Post}_i$. It would line up with the table discussed above as follows:

	Treatment	Control	Difference
Pre-period	.	β_0	β_1
Post-period	.	.	.
Difference	.	β_2	β_3